FINAL PROJECT REPORT

Internship Project Report

Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report

Submitted by

Priya Kumari

Submission Date : 30th July 2026

Final Project Report Template

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1: Introduction

1.1Project Overview

The project titled 'Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report' focuses on analyzing the adoption and impact of Artificial Intelligence (AI) in the education sector across different countries and regions. As AI technologies become increasingly integrated into teaching, learning, administration, assessment, and personalized education, there is a growing need to understand how different parts of the world are adopting these technologies. This project aims to transform raw educational AI adoption data into meaningful business intelligence using Tableau.

The project follows the complete Data Analytics lifecycle beginning with problem definition, project planning, data collection, data quality assessment, preprocessing, exploratory analysis, visualization, dashboard development, story creation, testing, and reporting. The dataset was cleaned and validated before analysis to ensure accuracy and consistency. Tableau was used to create interactive dashboards containing charts, maps, KPIs, filters, and stories that allow users to explore the data dynamically.

The analysis provides insights into AI adoption percentages, regional comparisons, country-wise performance, growth patterns, and key indicators influencing AI implementation in education. Interactive filters enable users to compare countries and regions easily and identify high-performing as well as low-performing areas.

The report is intended for policymakers, educational institutions, researchers, government bodies, technology providers, and students who require reliable, data-driven insights for strategic decision-making. The findings can help stakeholders understand current trends, identify gaps in AI adoption, and develop future strategies for improving AI integration in education.

1.2Objectives

The major objectives of this project are as follows:

1. To study the current status of AI adoption in education at the global level.

2. To identify regional and country-wise trends in AI implementation.

3. To clean, validate, and preprocess the dataset to improve data quality and reliability.

4. To perform exploratory data analysis for discovering hidden patterns and relationships.

5. To design effective Tableau visualizations that communicate complex information clearly.

6. To build interactive dashboards with filters and KPIs for easy analysis.

7. To develop a Tableau Story that presents insights in a logical sequence.

8. To compare AI adoption across different regions and educational environments.

9. To support evidence-based decision-making for governments, educational institutions, and researchers.

10. To provide recommendations based on analytical findings for improving AI adoption in education.

Project Initialization and Planning Phase

Date	21 July 2026
Team ID	Priya Kumari
Project Name	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Problem Statement (PS)	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	An Education Policy Maker	. Monitor AI adoption and usage in education across different countries.	It is difficult to compare AI adoption trends and identify areas that need irreverent.	The data is spread across multiple indicators and lacks a unified analysis platform.	Challenged in making effective policy decisions.
PS-2	A Student Counselor	Identify students who require psychological support	It is difficult to understand the relationship between stress, depression, and academic performance	There is no single visual platform showing all important mental health indicators	Challenged in providing early and effective counseling support

Initial Project Planning Template

Date	21july 2026
Team ID	Priya Kumari
Project Name	Global AI adoption in Education
Maximum Marks	4 Marks

Project Name: Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection	USN-1	As a user, I want to upload the AI usage dataset so that I can analyze educational trends.	2	High	1	20 july 2026	21 july 2026
Sprint-1	Data Preprocessing	USN-2	As a user, I want the system to clean and preprocess the dataset before analysis	2	High	1	20 july 2026	22 july 2026
Sprint-2	Dashboard	USN-3	As a user, I want to visualize AI usage using interactive charts and graphs.	3	High	1	23 july 2026	24 july 2026
Sprint-2	Data Analysis	USN-4	As a user, I want to compare AI adoption	3	Medium	1	24 july 2026	25 july 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
			across countries, regions, and education indicators.					
Sprint-3	Reports	USN-5	As a user, I want to generate summary reports from the dashboard for better decision-making.	2	High	1	26 July 2026	27 july 2026

Project Initialization and Planning Phase

Date	18 July 2026
Team ID	Priya kumari
Project Title	Global AI adoption in Education
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To develop, analyze, and visualize global Artificial Intelligence (AI) adoption trends in education using an interactive dashboard. The project aims to help educators, policymakers, and researchers understand AI usage among students and teachers across different countries through meaningful visualizations and data analytics.
Scope	The project includes cleaning and preprocessing the AI education dataset, performing exploratory data analysis, creating interactive dashboards using Tableau/Power BI, and deploying the dashboard through a web application developed with Python Flask. The project focuses on historical AI adoption trends and does not include real-time data collection or predictive machine learning models.
Problem Statement	
Description	The rapid adoption of Artificial Intelligence in education has created a large amount of educational data across different countries. However, these datasets are often stored in spreadsheets and are difficult to interpret. Decision-makers require an interactive platform that clearly presents AI adoption patterns, usage statistics, government policies, and educational trends.
Impact	Providing an interactive analytics platform will enable educators, researchers, and policymakers to better understand global AI adoption in education. The insights can support policy formulation, curriculum planning, technology

	investments, and educational research.
Proposed Solution	
Approach	The project follows a structured data analytics workflow. The dataset will first be cleaned and preprocessed using Python. Exploratory Data Analysis (EDA) will be performed to identify trends and patterns. Interactive dashboards will then be created using Tableau or Power BI to visualize AI adoption across countries, regions, and educational sectors. Finally, the dashboards will be embedded into a Flask-based web application and deployed using Render for public access.
Key Features	• **Interactive dashboard displaying AI adoption across countries.** • **Country-wise comparison of student and teacher AI usage.** • **Visualization of AI adoption in schools over different months and years.** • **Analysis of government AI policies and curriculum integration.** • **Urban versus rural AI usage comparison.** • **Gender gap analysis in AI usage.** • **Top AI tools used in different countries.** • **Internet penetration and education index analysis.** • **Responsive web interface for easy accessibility.** • **Cloud deployment using Render and GitHub.**

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Computing Resources	Personal Computer/Laptop
Memory	Processor	Intel Core i5/Ryzen 5 or above
Storage	Memory	Minimum 8 GB RAM
Software		

Frameworks	Programming Language	Python 3.12
Libraries	Additional libraries	Gunicorn (v21.2.0) for stable cloud routing execution
Development Environment	IDE, version control	VS Code (or Text Editor), macOS Terminal, Git, and GitHub UI
Data		
Data	Data set	Global AI Usage in Education Dataset (CSV Format)

Data Collection and Preprocessing Phase

Date	22 July 2026
Team ID	Priya Kumari
Project Title	Global AI adoption in Education
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	**Description**
Data Overview	The dataset contains global AI adoption statistics in education across multiple countries and months. It includes information on student AI usage, teacher AI usage, school AI adoption, average daily AI usage, top AI tools, internet penetration, education index, government AI policies, urban and rural AI usage, AI curriculum integration, and gender gap in AI usage.
Data Cleaning	**Missing Values:** Identified missing values in columns such as Government AI Policy and replaced them with appropriate values (e.g., "Unknown") or removed incomplete records where necessary. **Duplicates:** Checked and removed duplicate records to maintain data integrity. **Error Correction:** Corrected inconsistent spellings, removed extra spaces, standardized country and region names, and verified percentage values were within valid ranges (0–100).
Data Transformation	**Calculated Fields:** Created new calculated metrics such as Total AI Usage, Urban-Rural Usage Difference, and Student-Teacher Usage Ratio. **Aggregations & Pivot Tables:** Used Pivot Tables and summary statistics to analyze AI adoption by country, region, and year. **Filtering:** Removed invalid or extreme values that could negatively affect the analysis and retained only meaningful observations.
Data Type Conversion	Converted Year and Month columns to integer format, percentage-related columns to numeric data types, Education Index to decimal format, Government AI Policy and AI in Curriculum to categorical variables, and

	Country, Region, and Top AI Tool to text data types for accurate visualization and analysis.
Column Splitting and Merging	Standardized country and region information where necessary. Combined Year and Month columns into a Date field for time-series analysis. Created additional derived columns such as AI Adoption Category and Internet Accessibility Level for improved dashboard visualization.
Data Modeling	Organized the cleaned dataset into a structured format suitable for Tableau/Power BI. Established relationships between geographical, educational, and AI adoption attributes to support comparative analysis across countries and regions. Calculated summary measures such as average AI usage, regional adoption rates, and curriculum implementation percentages.
Save Processed Data	Saved the cleaned and transformed dataset as a CSV/Excel file with standardized column names and formats. Verified the dataset before importing it into Tableau/Power BI and ensured all columns were correctly recognized for visualization.

Data Collection and Preprocessing Phase

Date	25 July 2026
Team ID	Priya Kumari
Project Title	Global AI adoption in Education
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	**Data Quality Issue**	**Severity**	**Resolution Plan**
Global AI Usage in Education Dataset (CSV)	Missing values in fields such as **Government AI Policy** and other optional attributes.	Moderate	Replace missing values with appropriate labels (e.g., "Unknown") or remove incomplete records where necessary.
Global AI Usage in Education Dataset (CSV)	Inconsistent text formatting, including extra spaces, inconsistent capitalization, and spelling variations in categorical columns (Country, Region, Government AI Policy, Top AI Tool).	Moderate	Trim whitespace, standardize capitalization, correct spelling inconsistencies, and normalize text values before analysis.

Global AI Usage in Education Dataset (CSV)	Duplicate records that may affect statistical analysis and dashboard accuracy.	High	Identify duplicate rows using unique combinations of Year, Month, and Country, and remove redundant records.
Global AI Usage in Education Dataset (CSV)	Invalid or out-of-range percentage values in AI usage and internet penetration columns.	High	Validate that all percentage values fall within the acceptable range (0–100%) and correct or remove invalid entries.
Global AI Usage in Education Dataset (CSV)	Incorrect data types for numerical and categorical columns.	Low	Convert Year and Month to integers, percentage columns to numeric format, Education Index to decimal, and categorical fields to text.
Global AI Usage in Education Dataset (CSV)	Inconsistent date representation due to separate Year and Month columns.	Low	Merge Year and Month into a single Date field for easier time-series analysis and visualization.

Data Collection and Preprocessing Phase

Date	22 July 2026
Team ID	Priya Kumari
Project Title	Global AI adoption in Education
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	**Description**
Data Overview	The dataset contains global AI adoption statistics in education across multiple countries and months. It includes information on student AI usage, teacher AI usage, school AI adoption, average daily AI usage, top AI tools, internet penetration, education index, government AI policies, urban and rural AI usage, AI curriculum integration, and gender gap in AI usage.
Data Cleaning	**Missing Values:** Identified missing values in columns such as Government AI Policy and replaced them with appropriate values (e.g., "Unknown") or removed incomplete records where necessary. **Duplicates:** Checked and removed duplicate records to maintain data integrity. **Error Correction:** Corrected inconsistent spellings, removed extra spaces, standardized country and region names, and verified percentage values were within valid ranges (0–100).
Data Transformation	**Calculated Fields:** Created new calculated metrics such as Total AI Usage, Urban-Rural Usage Difference, and Student-Teacher Usage Ratio. **Aggregations & Pivot Tables:** Used Pivot Tables and summary statistics to analyze AI adoption by country, region, and year. **Filtering:** Removed invalid or extreme values that could negatively affect the analysis and retained only meaningful observations.
Data Type Conversion	Converted Year and Month columns to integer format, percentage-related columns to numeric data types, Education Index to decimal format, Government AI Policy and AI in Curriculum to categorical variables, and Country, Region, and Top AI Tool to text data types for accurate visualization

	and analysis.
Column Splitting and Merging	Standardized country and region information where necessary. Combined Year and Month columns into a Date field for time-series analysis. Created additional derived columns such as AI Adoption Category and Internet Accessibility Level for improved dashboard visualization.
Data Modeling	Organized the cleaned dataset into a structured format suitable for Tableau/Power BI. Established relationships between geographical, educational, and AI adoption attributes to support comparative analysis across countries and regions. Calculated summary measures such as average AI usage, regional adoption rates, and curriculum implementation percentages.
Save Processed Data	Saved the cleaned and transformed dataset as a CSV/Excel file with standardized column names and formats. Verified the dataset before importing it into Tableau/Power BI and ensured all columns were correctly recognized for visualization.

Business Question and Visualization Report

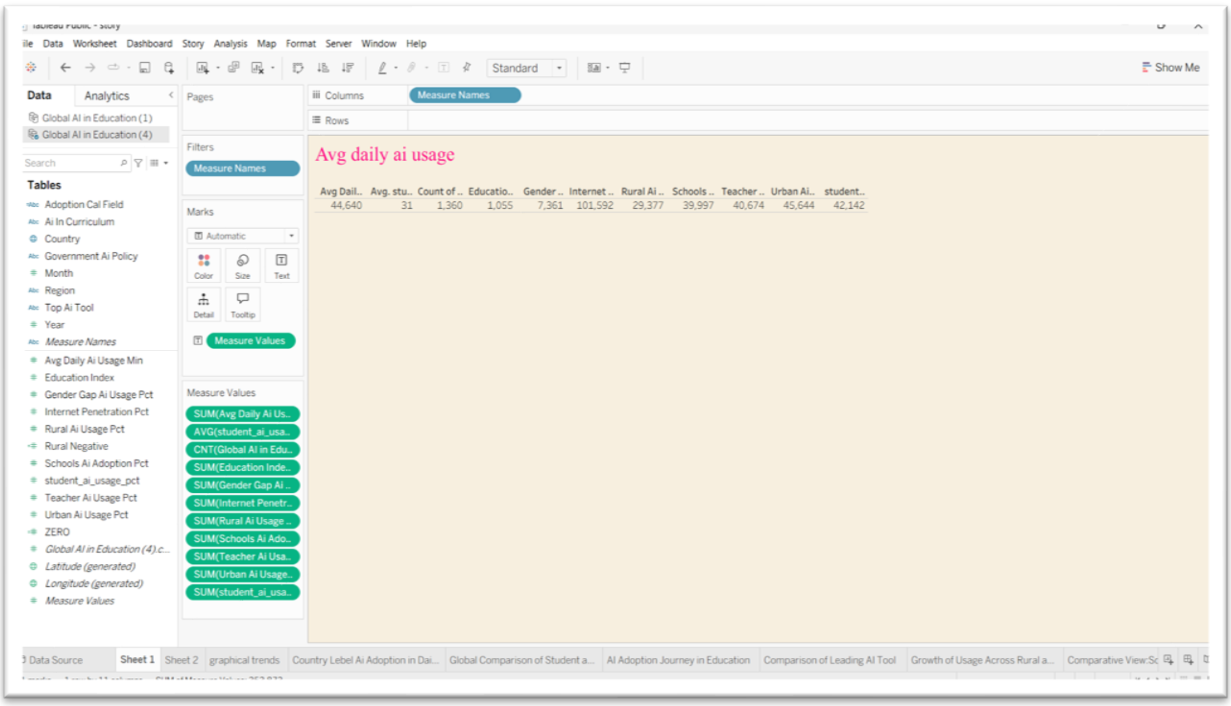
Date	22 July 2026
Team ID	Priya Kumari
Project Name	Global AI adoption in Education
Maximum Marks	5 Marks

Business Questions and Visualizations

Q1. What are the key performance indicators that summarize global AI adoption in education?

- Sheet Number: Sheet 1
- Visualization: KPI Cards / Text Table
- Insight Provided: Provides an overall snapshot of AI adoption, internet penetration,

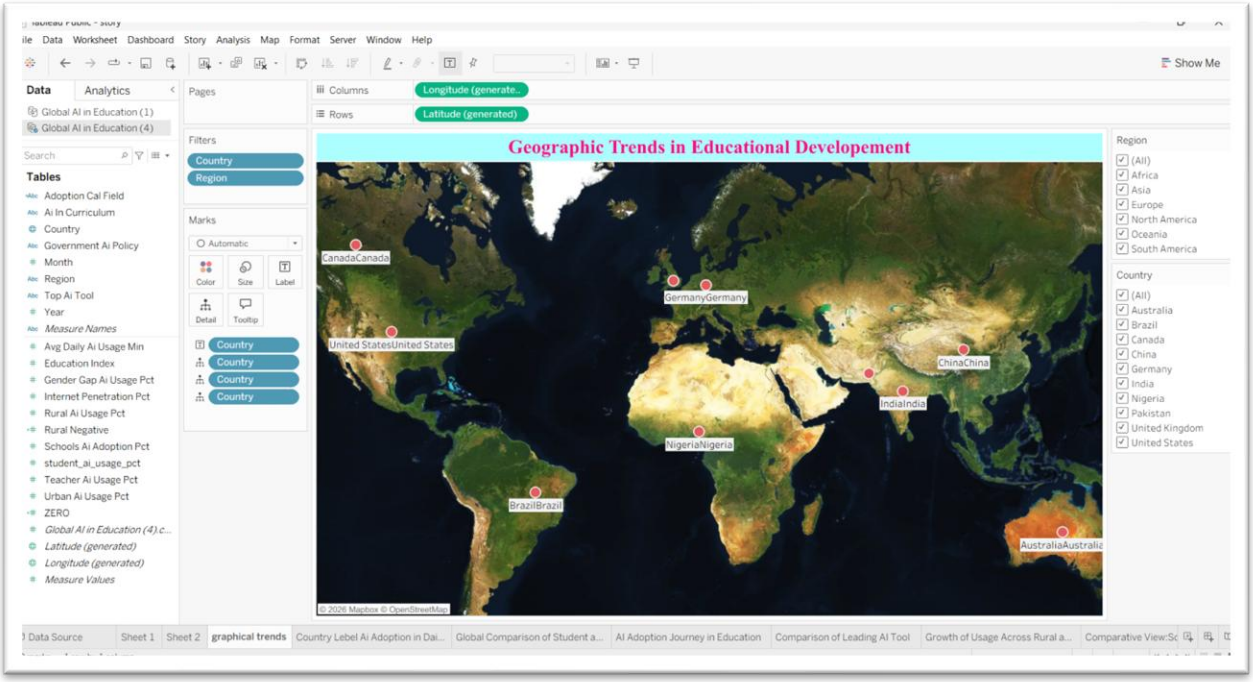
school AI adoption, student and teacher AI usage, and urban/rural AI usage



Q2. How is AI adoption in education distributed across different countries and regions?

- Visualization: Geographical Symbol Map

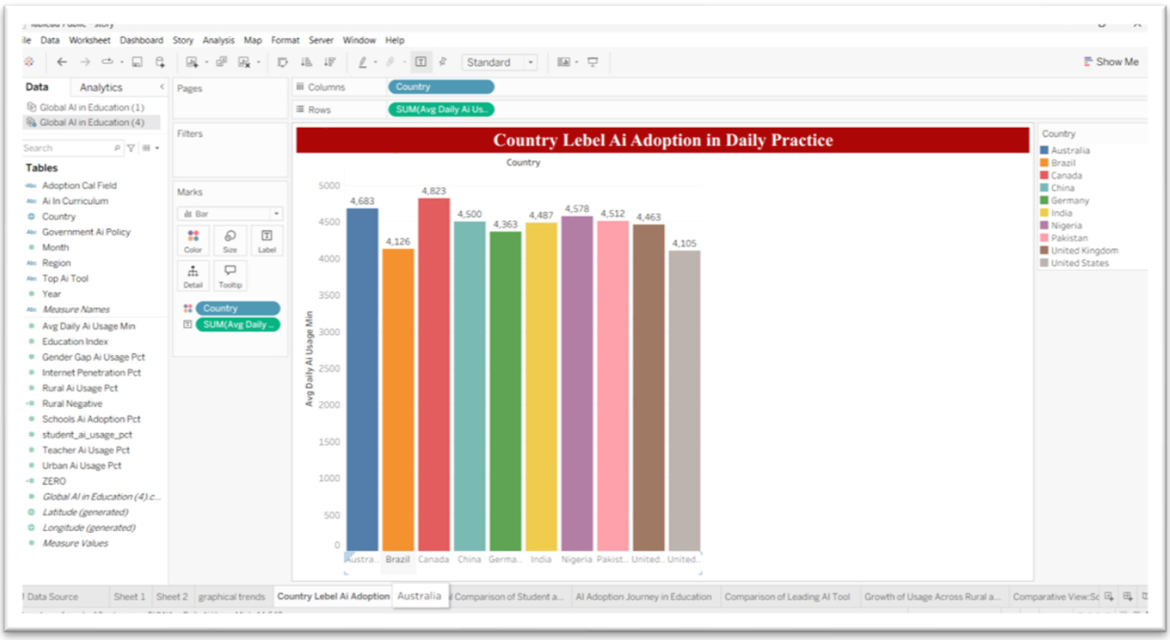
• Insight Provided: Displays the geographical distribution of AI adoption across countries and regions.



Q3. Which country has the highest average daily AI usage in education?

• Visualization: Vertical Bar Chart

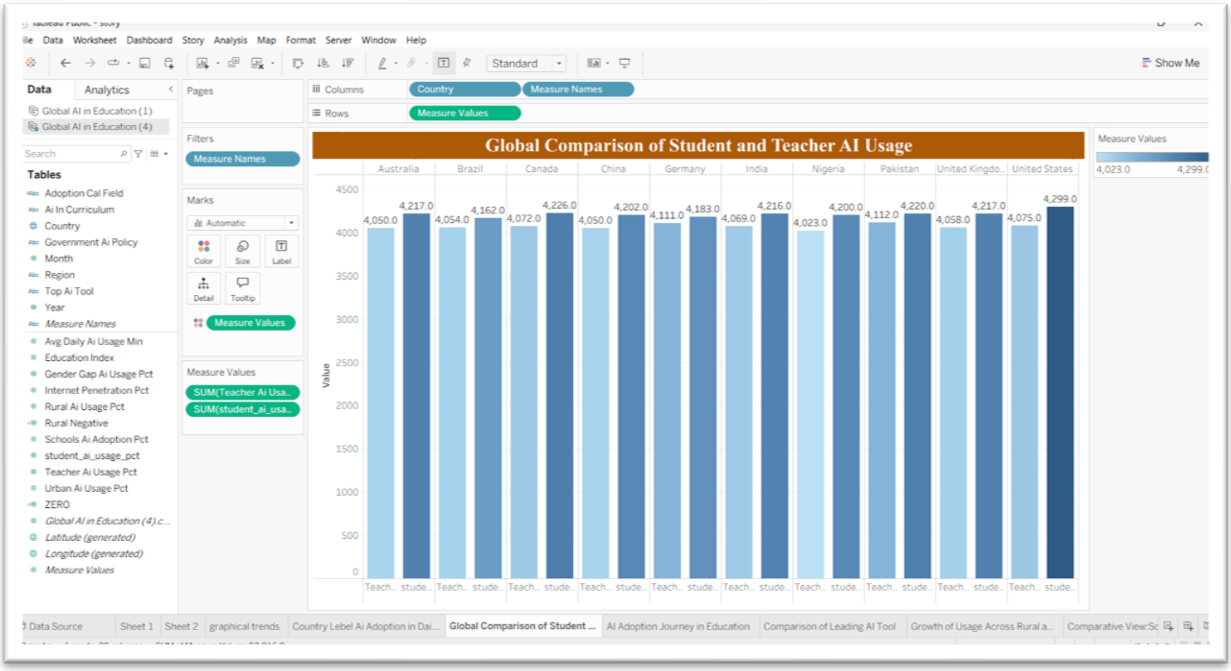
• Insight Provided: Compares average daily AI usage among countries.



Q4. How does student AI usage compare with teacher AI usage across countries?

• Visualization: Side-by-Side (Clustered) Bar Chart

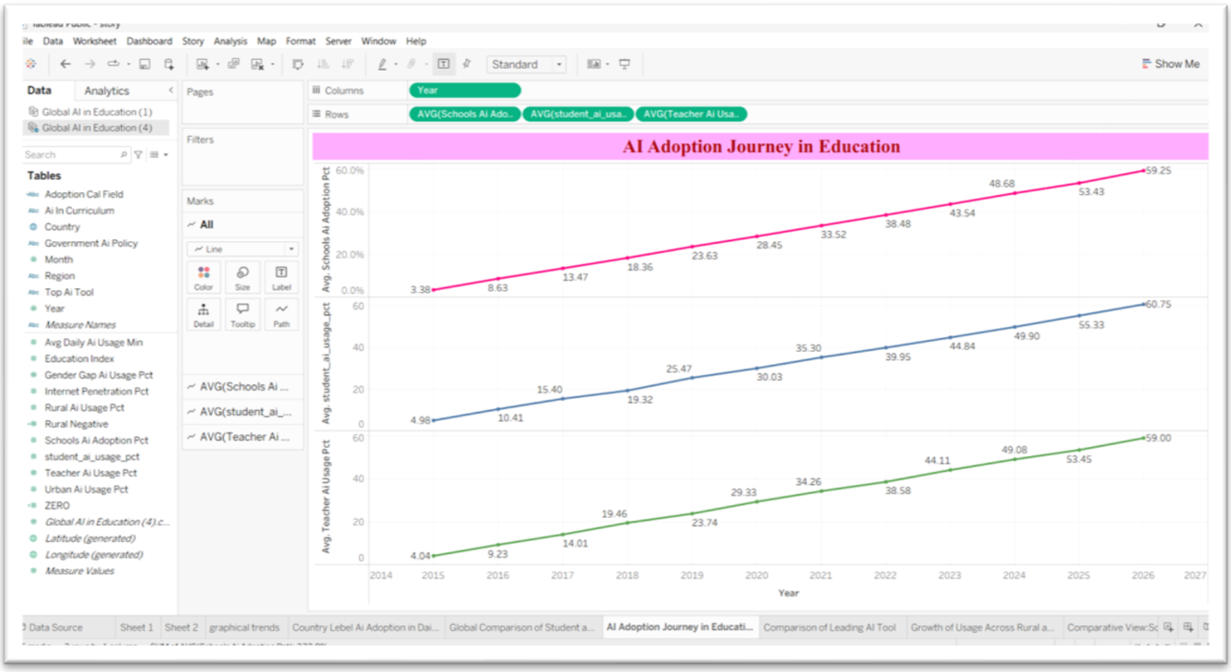
• Insight Provided: Compares average AI usage by students and teachers for each country.



Q5. How has AI adoption in education evolved from 2014 to 2026?

• Visualization: Multi-Line Chart

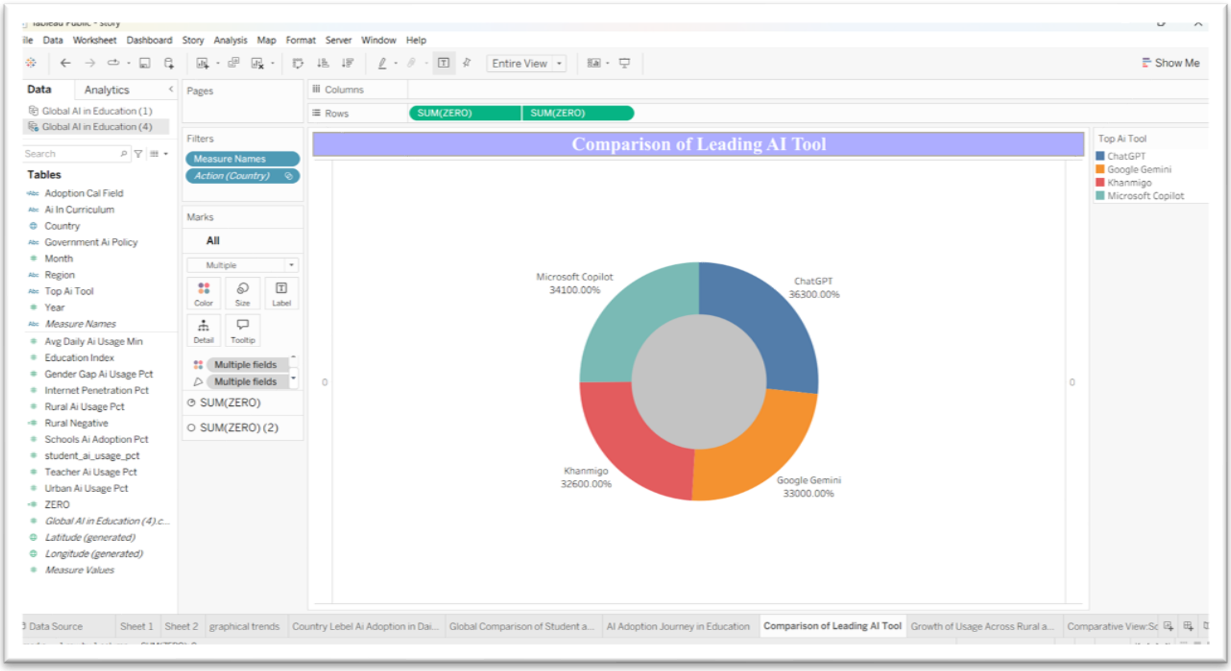
• Insight Provided: Shows yearly trends in school, student and teacher AI adoption.



Q6. Which AI tools are most commonly used in education?

•Visualization: Donut Chart

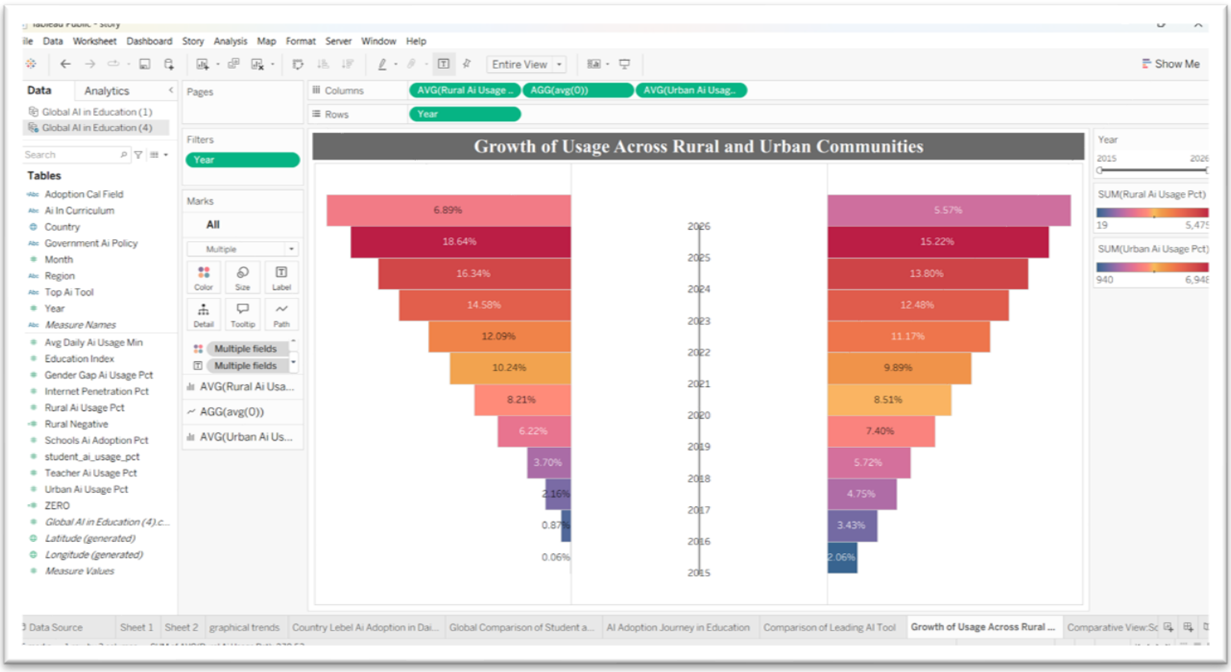
• Insight Provided: Shows the share of ChatGPT, Google Gemini, Khanmigo and Microsoft Copilot.



Q7. How has AI usage grown in rural versus urban communities over time?

•Visualization: Butterfly Butterfly Bar Chart

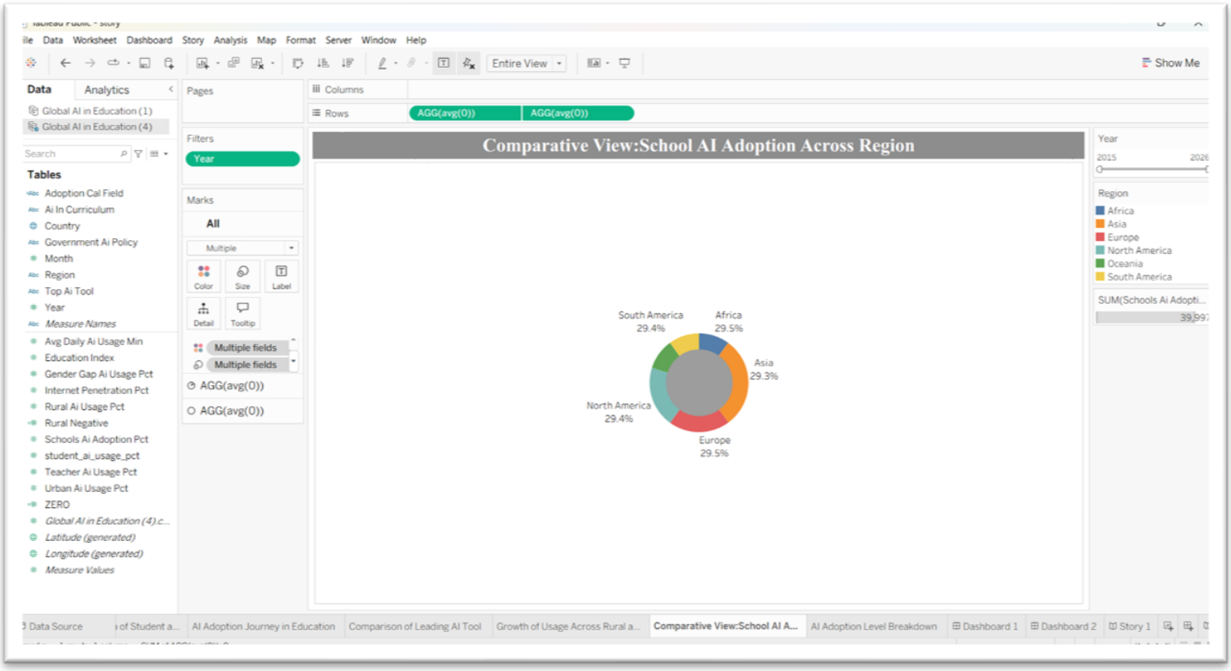
• Insight Provided: Compares rural and urban AI usage growth year by year.



Q8. How does school AI adoption differ across global regions?

• Visualization: Donut Chart

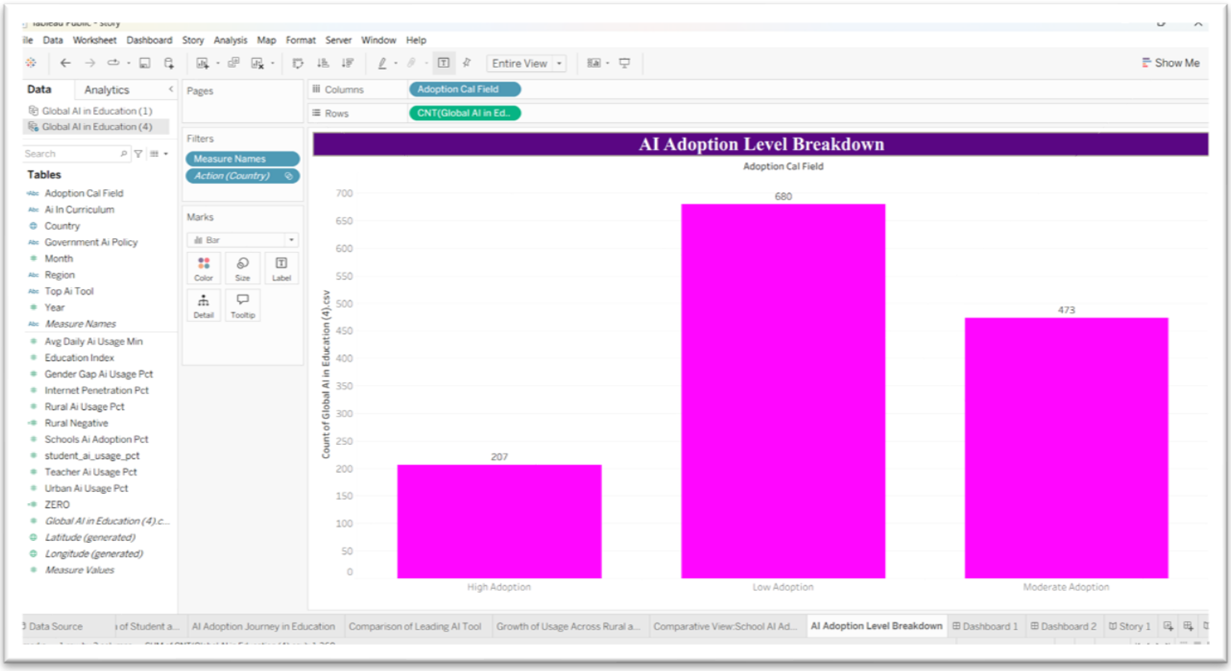
• Insight Provided: Compares school AI adoption across continents/regions.



Q9. What is the distribution of institutions by AI adoption level?

• Visualization: Vertical Bar Chart

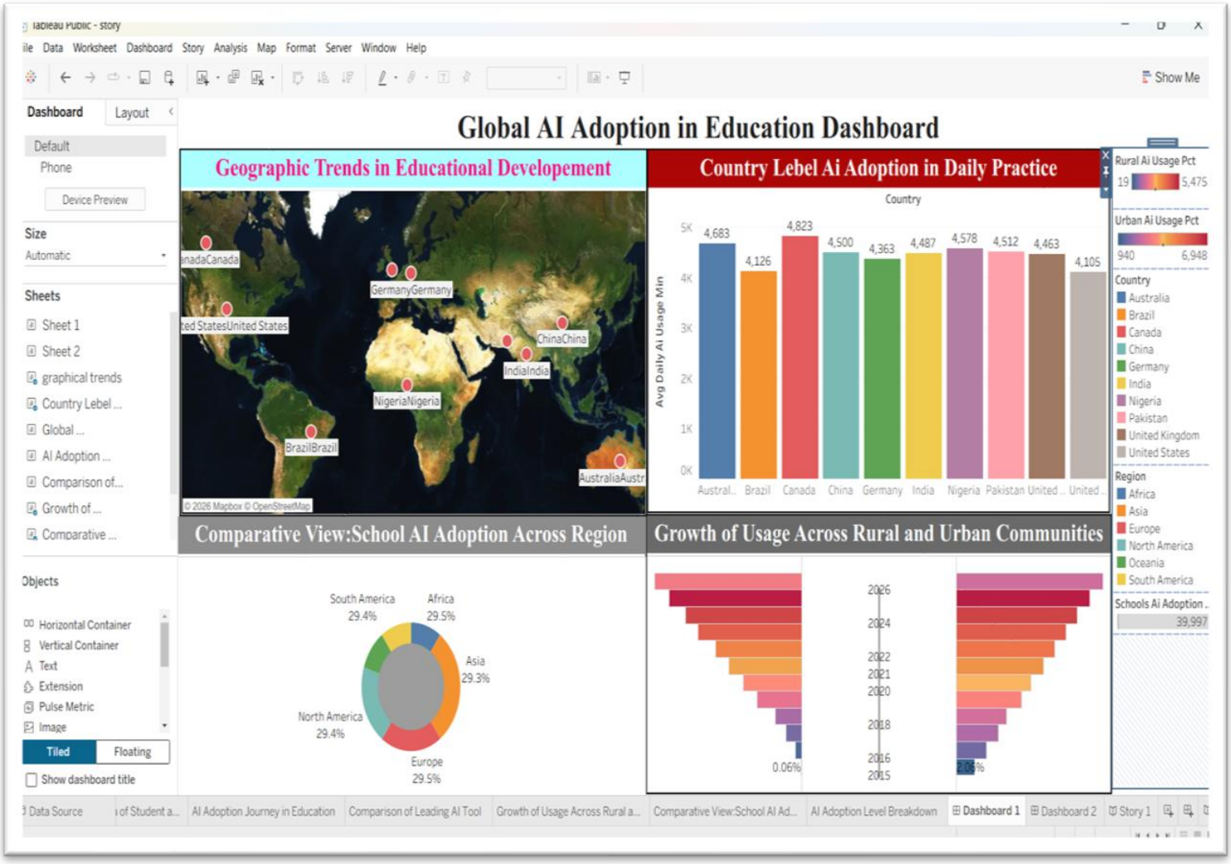
• Insight Provided: Shows High, Moderate and Low AI adoption categories.

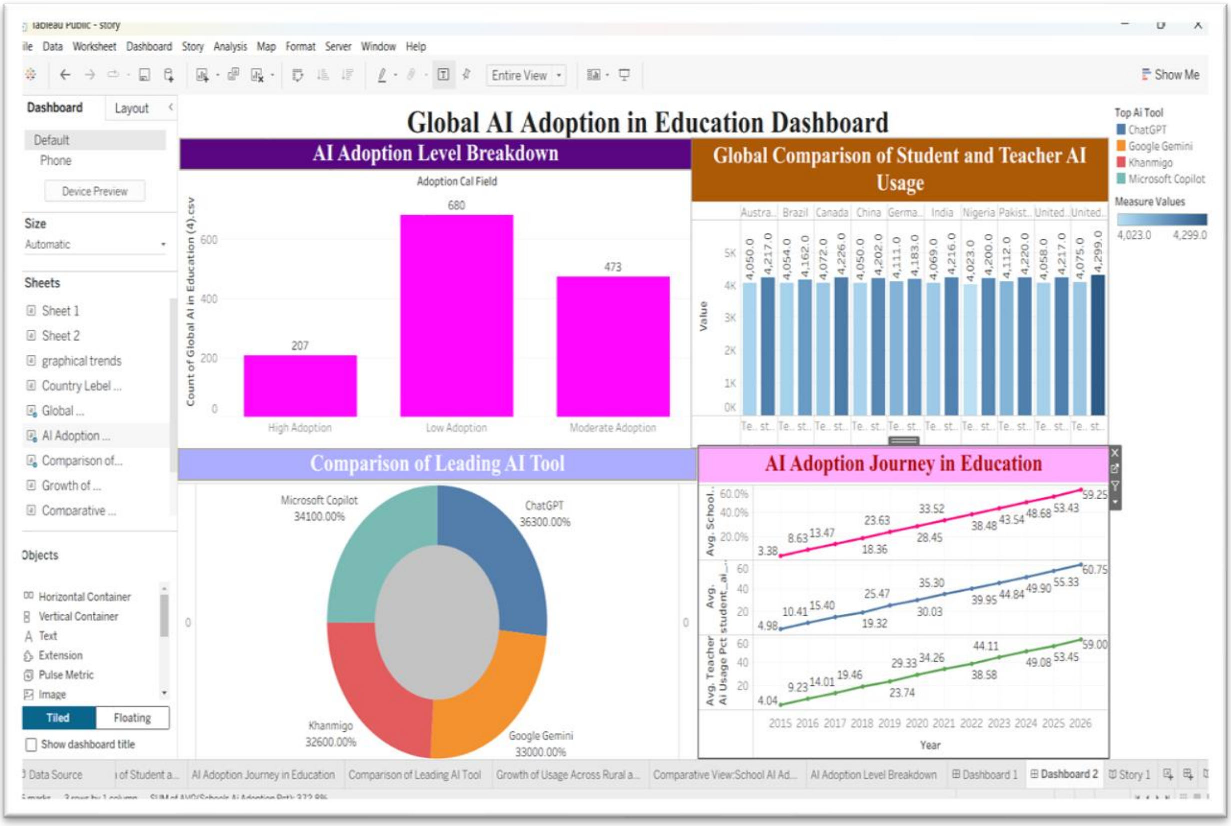


Dashboard Design

Date	24 July 2026
Team ID	Priya Kumari
Project Name	Global AI adoption in Education
Maximum Marks	5 Marks

Activity 1: Interactive and Visually Appealing Dashboards





The dashboard has been designed by following key visualization principles such as clear layout, consistent color theme, interactive filters, user-friendly navigation, and responsive design. The Tableau dashboard enables users to explore global AI adoption in education through multiple interactive visualizations and gain meaningful insights from the dataset.

Key Insights from the Dashboard

- Global AI Adoption Overview: The dashboard presents a comprehensive overview of AI adoption in education across different countries and regions, highlighting global adoption patterns and educational development trends.

- Country-wise AI Usage: Canada records the highest average daily AI usage, while other countries such as Australia, China, India, Germany, Nigeria, Pakistan, Brazil, the United Kingdom, and the United States show varying adoption levels, reflecting differences in AI implementation.

- Regional Comparison: AI adoption is distributed almost equally across Asia, Europe, Africa, North America, and South America, indicating that AI is gaining widespread acceptance in education worldwide.

- AI Adoption Levels: Most educational institutions fall under the Low Adoption category, followed by Moderate Adoption, while High Adoption has the smallest share, suggesting considerable scope for future AI expansion.

- Leading AI Tools: ChatGPT, Google Gemini, Microsoft Copilot, and Khanmigo are among the most widely used AI tools, demonstrating the growing role of generative AI in modern education.

- Student and Teacher AI Usage: AI usage by both students and teachers has increased steadily from 2015 to 2026, showing continuous growth in AI integration within teaching and learning processes.

- Rural and Urban Growth: AI adoption has grown consistently in both rural and urban communities, reflecting improved digital infrastructure and greater accessibility to AI technologies.

- Interactive Analysis & Decision Support: The dashboard enables users to analyse AI adoption through interactive filters such as Country, Region, AI Tool, and Adoption Level, helping educators, researchers, and policymakers identify trends, compare performance, and make informed decisions for future AI implementation.

Activity 2: Live Deployment Link

Below is the verified public access link to the interactive, cloud-hosted visualization story:

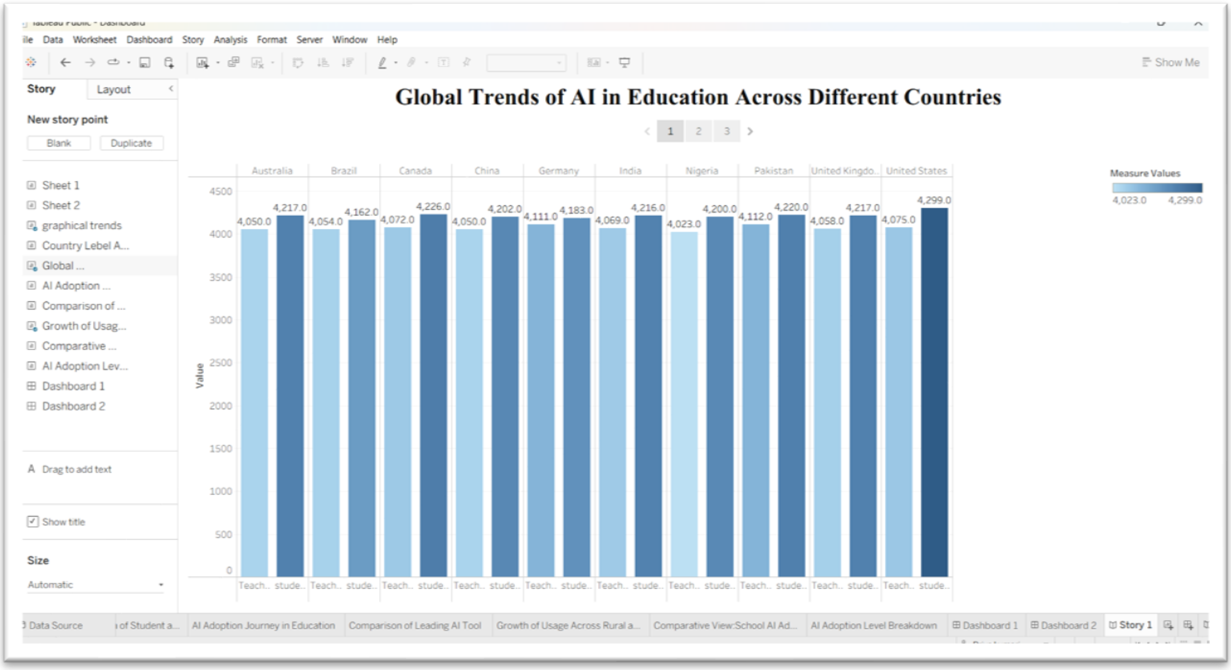
Dashboard Tableau Public Link

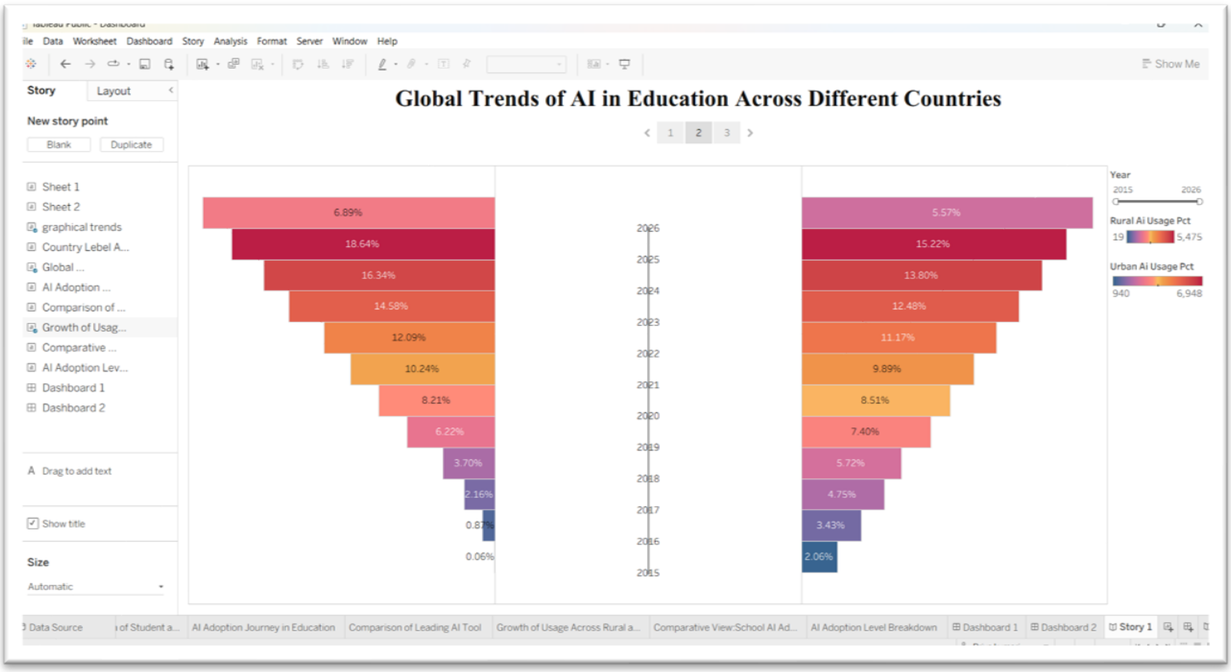
https://public.tableau.com/views/dashboard1_17851507810350/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

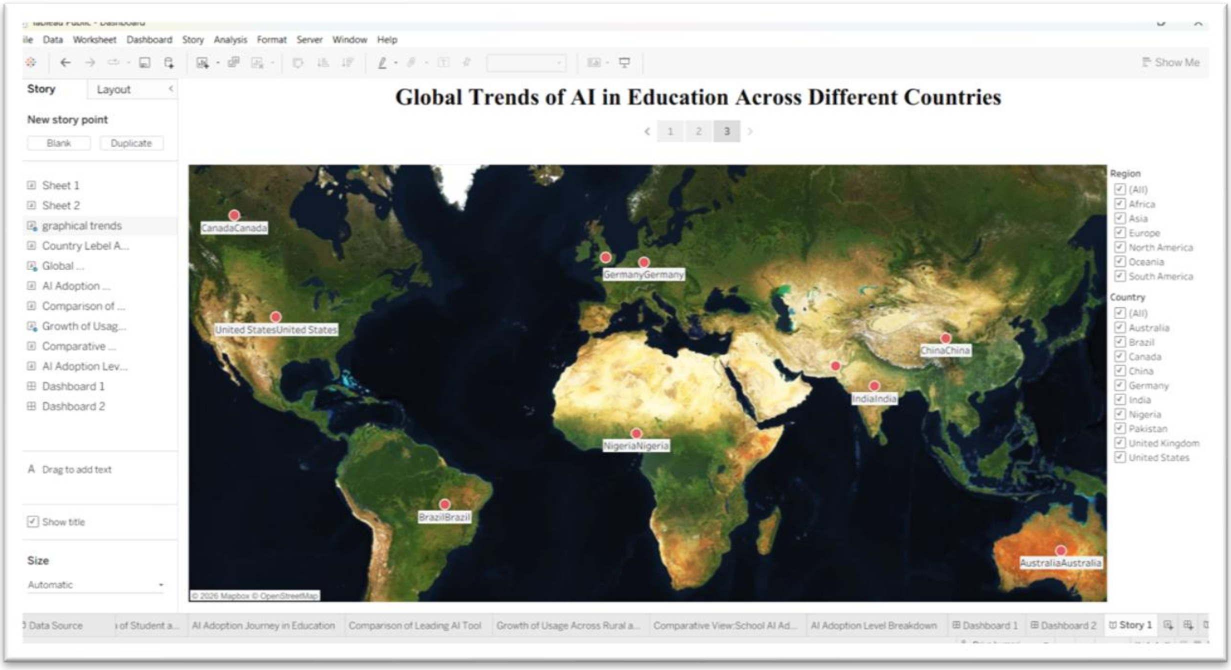
Story

Date	24 July 2026
Team ID	Priya kumari
Project Name	Global AI Adoption in Education
Maximum Marks	5 Marks

Activity 1: Story Design:







Activity 2: Publish Story on Tableau Public and Paste Public link below

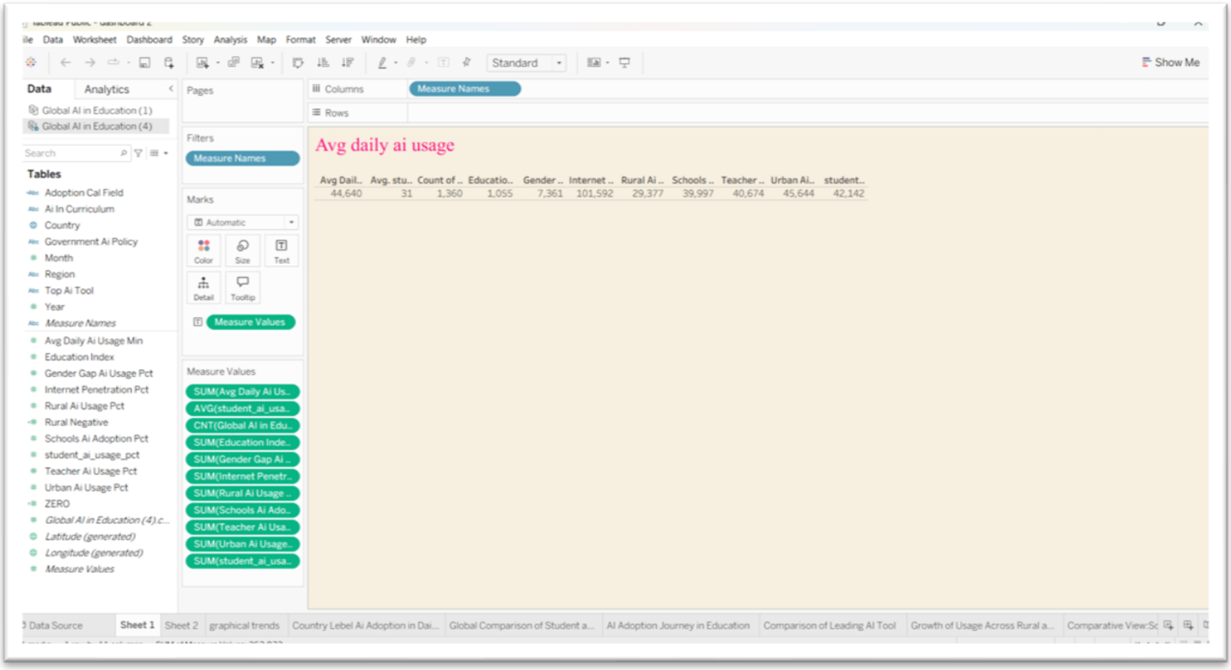
https://public.tableau.com/shared/T7SRJDYQB?:display_count=n&:origin=viz_share_link

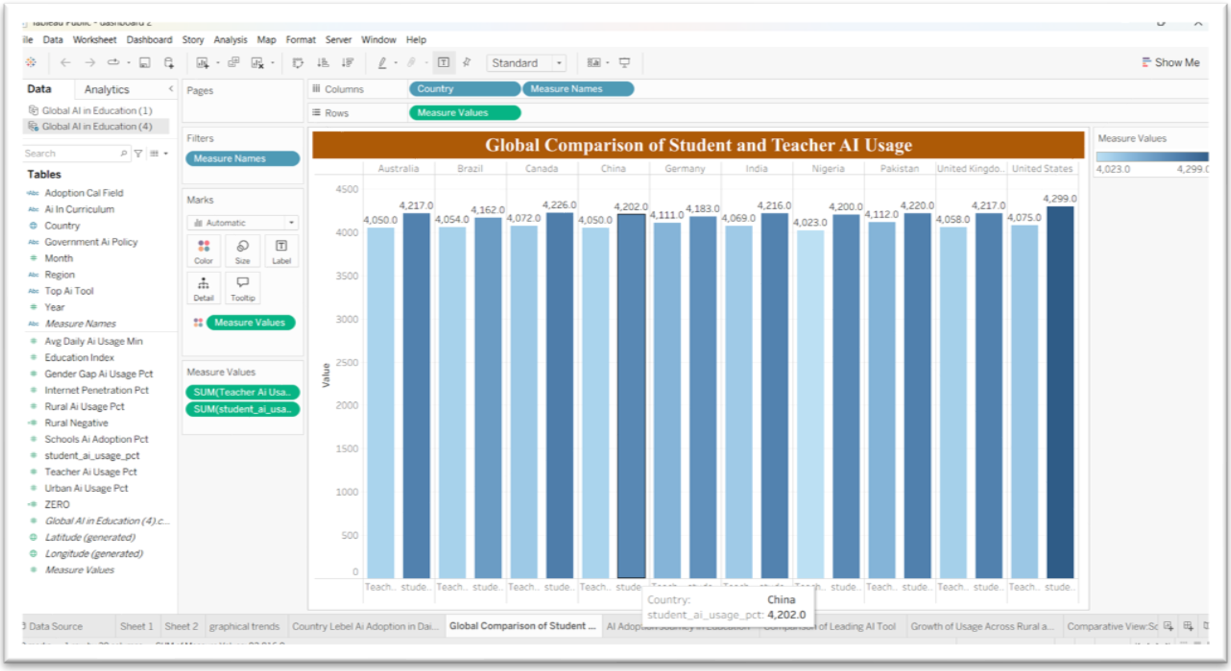
7. Performance Testing

The workbook was tested to ensure correct filter behavior, accurate calculated fields, responsive dashboard interaction, and consistent visualization output.

7.1 Utilization of Data Filters

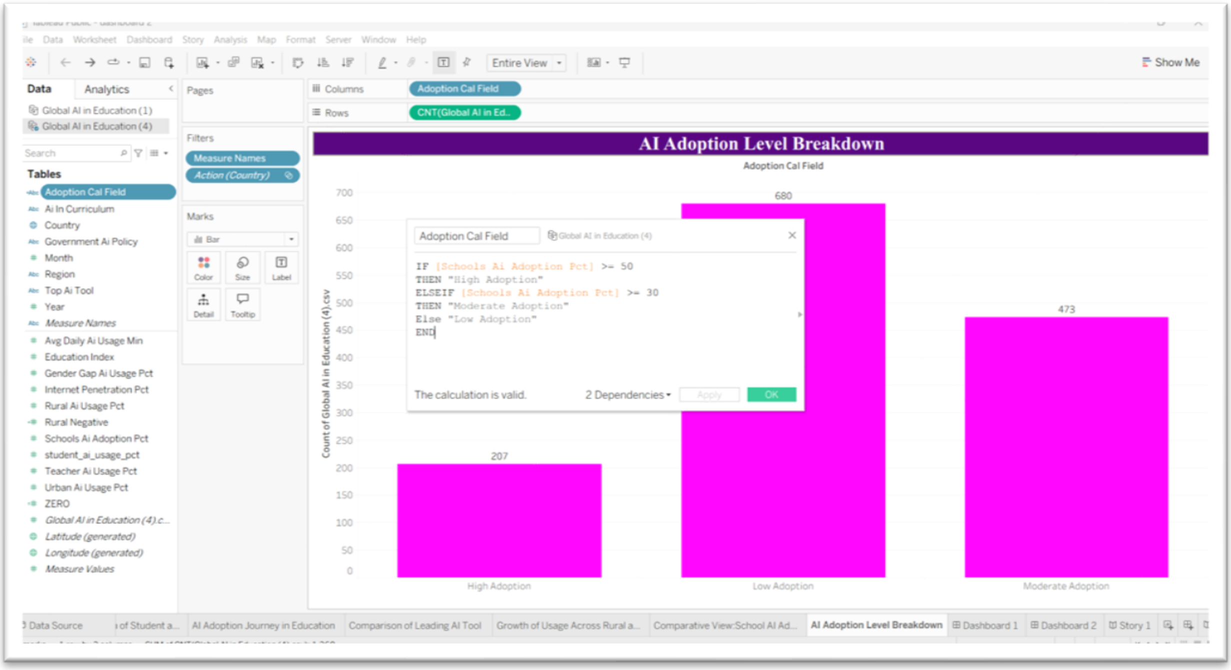
Interactive filters such as Region, Country, Year, AI Category, and other dataset fields improve analysis.





7.2 Number of Calculation Fields

Calculated fields were created wherever required for KPIs, percentages, and derived metrics.



7.3 Number of Visualizations

The workbook contains 9 worksheets, 2 dashboards, and 1 story.

8. Conclusion/Observation

The project "Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report" successfully analyzed global AI adoption trends in the education sector using Tableau. Through systematic data collection, preprocessing, visualization, dashboard development, and story presentation, meaningful insights were generated from the available dataset.

The analysis revealed differences in AI adoption across countries and regions, highlighting areas with strong implementation as well as regions where AI adoption remains comparatively low. Interactive dashboards simplified complex information and enabled users to compare performance efficiently.

The project also demonstrated the importance of data visualization in transforming raw data into actionable insights. Tableau's interactive features—including dashboards, filters, KPIs, maps, and stories—made the analysis more effective and user-friendly.

The project enhanced practical knowledge of data analytics, data preprocessing, dashboard design, visualization principles, and business intelligence reporting. It also improved analytical thinking, problem-solving skills, and the ability to communicate insights through visual storytelling.

Overall, the project achieved all its objectives and provides a valuable analytical tool for educators, researchers, policymakers, and institutions interested in understanding global AI adoption in education

9. Future Scope

9.1 Recommendations

Based on the analysis, the following recommendations are suggested:

Educational institutions should increase investment in AI-powered learning technologies.

Governments should develop national AI education strategies to encourage wider adoption.

Teacher training programs should include AI literacy and digital competency development.

Institutions should regularly monitor AI adoption using data-driven dashboards.

Data collection should be standardized across countries to improve future comparisons.

Organizations should promote responsible and ethical use of Artificial Intelligence in education.

Decision-makers should use business intelligence dashboards for continuous monitoring and policy formulation.

Educational institutions should periodically evaluate AI implementation outcomes to improve effectiveness.

9.2 Future Scope

Although the project provides comprehensive insights, there are several opportunities for future enhancement.

Future versions of the project may include:

Real-time educational AI datasets.

Predictive analytics using Machine Learning models.

Time-series analysis to study AI adoption growth over multiple years.

Student performance analysis before and after AI implementation.

Institution-level comparisons.

Integration with Power BI, Python, SQL, and cloud-based databases.

Automated dashboard refresh using Tableau Server or Tableau Cloud.

Advanced forecasting models to estimate future AI adoption.

Inclusion of additional educational indicators such as digital infrastructure, internet accessibility, government expenditure, and teacher readiness.

AI-powered recommendation systems for policymakers and educational institutions.

These improvements would further increase the practical value of the project and support better strategic decision-making.

10. Appendix

Dataset: CSV file:

https://drive.google.com/file/d/1m-BpmDp9oWII4NeIzuKlg__axJZZJTuW/view?usp=sharing

GitHub Repository:

https://github.com/priyakumari2k6-lgtm/Tableau-Public/tree/main

Tableau Public Project Link:

https://public.tableau.com/app/profile/priya.kumari1471/vizzes

Project Demo Video

https://drive.google.com/file/d/10WeGkmiZhG-vJ6sIkIf3VmhztZOsJgzZ/view?usp=drivesdk